2 **(a)** Since *k* and *m* are constant, $a \propto -x$.
This implies that the block's acceleration is proportional to its displacement from the equilibrium position.

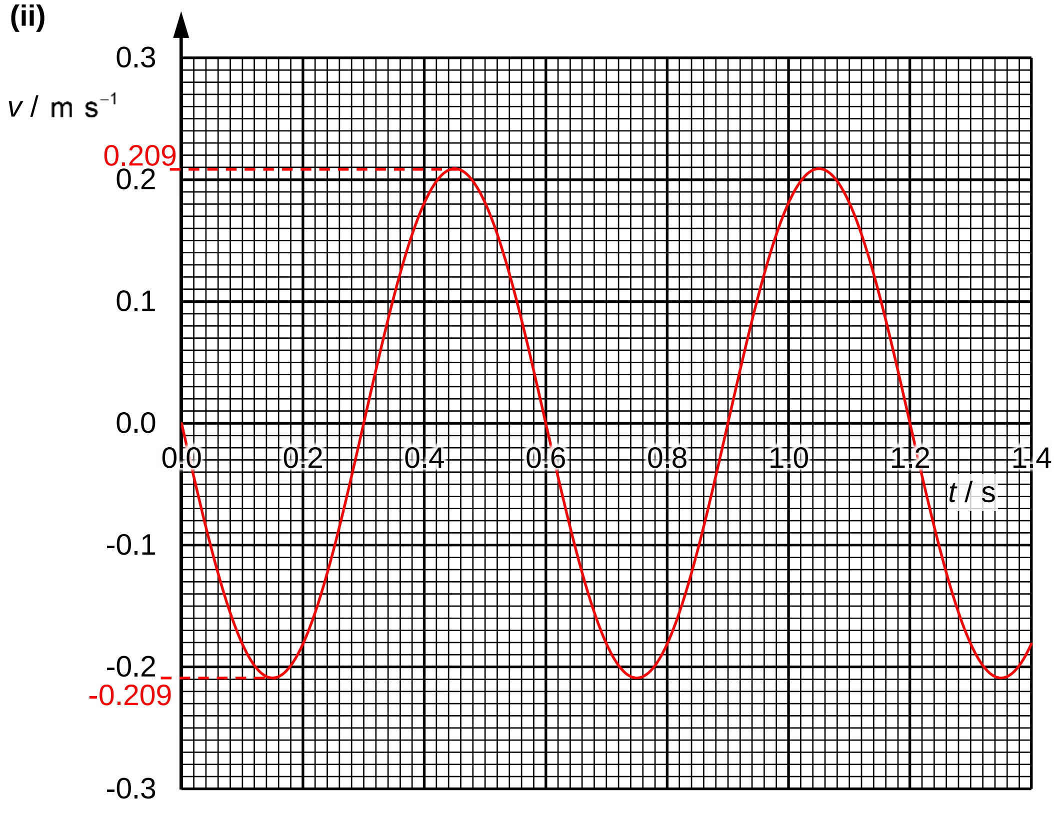
The negative sign implies that the direction of its acceleration is always opposite to its displacement, pointing towards the equilibrium position.

This satisfies the definition for simple harmonic motion.

(b) **(i)** From the graph, $T = 0.60$ s, $x_0 = 2.0 \times 10^{-2}$ m

$$\begin{aligned} v_0 &= \omega x_0 \\ &= \frac{2\pi}{T} x_0 \\ &= \frac{2\pi}{0.60}\left(2.0 \times 10^{-2}\right) \\ &= 0.20944 = 0.209 \text{ m s}^{-1} \end{aligned}$$

(ii)



(iii) $E_K = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\pm\omega\sqrt{{x_0}^2 - x^2}\right)^2 = \frac{1}{2}m\omega^2\left({x_0}^2 - x^2\right)$

$E_P = E_T - E_K = \frac{1}{2}m\omega^2{x_0}^2 - \frac{1}{2}m\omega^2\left({x_0}^2 - x^2\right) = \frac{1}{2}m\omega^2x^2$

$$E_P = E_K$$

$$\frac{1}{2}m\omega^2x^2 = \frac{1}{2}m\omega^2\left({x_0}^2 - x^2\right)$$

$$2x^2 = {x_0}^2$$

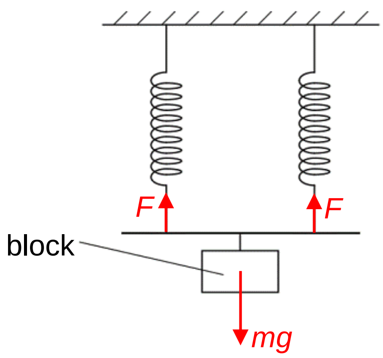
$$x = \pm\frac{x_0}{\sqrt{2}} = \pm\frac{2.0\times10^{-2}}{\sqrt{2}} = \pm0.014142\text{ m} = \pm1.4142\text{ cm}$$

At equilibrium, $L = 16.0\text{ cm}$.
When $E_P = E_K$,
$L = 16.0 + 1.4142 = 17.4142 = 17.4\text{ cm}$ (block is below equilibrium)
OR
$L = 16.0 - 1.4142 = 14.5858 = 14.6\text{ cm}$ (block is above equilibrium)

(c) $F_{eff} = 2F = mg$

$k_{eff}e = 2ke = mg$

$k_{eff} = 2k$



The <u>same mass results in half the extension</u> in each spring at equilibrium compared to a single spring in Fig. 2.1. Hence the <u>effective force constant is twice</u> the force constant of one spring ($k_{eff} = 2k$).

From $a = -\frac{k}{m}x$, angular frequency $\omega = \frac{2\pi}{T} = \sqrt{\frac{k}{m}}$. Hence period $\underline{T = 2\pi\sqrt{\frac{m}{k}}}$.

When the force constant is twice that in (a), the <u>period decreases to $\frac{1}{\sqrt{2}}$ the period in (a)</u>.

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3 **(a)** amplitudes: